

Kamal Yeshodhar Shastry Gattu

Greater Boston, USA | 571-732-7250 | kysgattu0502@gmail.com | [LinkedIn.com/in/kysgattu](https://www.linkedin.com/in/kysgattu) | [Github.com/kysgattu](https://github.com/kysgattu)

PROFESSIONAL SUMMARY

Software systems engineer with 4+ years of experience across software development, machine learning, and data engineering. Currently lead a small engineering team at a preclinical research laboratory, owning a portfolio of 30+ internal tools spanning computer vision, workflow automation, IoT monitoring, data security, and infrastructure. Take projects end to end, from requirements gathering through deployment and long-term support, with disciplined version control and DEV/PROD practices. Hold an M.S. in Computer Science with a strong foundation in computer vision, natural language processing, and enterprise ETL.

EDUCATION

Master of Science in Computer Science

Dec 2023

University of Massachusetts Lowell, Lowell, MA; GPA: 3.736/4

Bachelor of Technology in Computer Science & Engineering

Sep 2020

Jawaharlal Nehru Technological University Hyderabad, Hyderabad, India; GPA: 6.73/10

TECHNICAL SKILLS

Programming Languages: *Python, Java, C, C++, C#, HTML/CSS, JavaScript, Google Apps Script, PowerShell, Bash*

Databases: *PL/SQL, T-SQL, Oracle SQL, MySQL, MariaDB, Microsoft SQL Server, PostgreSQL, Amazon Redshift*

Python Libraries: *NumPy, Pandas, Scikit-learn, Keras, TensorFlow, PyTorch, NLTK, OpenCV, Matplotlib, Tkinter, gspread, Apache Tika*

Application Deployment Tools: *Flask, Docker, Streamlit, PyInstaller*

Machine Learning and AI: *Image Processing, Computer Vision, Natural Language Processing, Deep Learning, Transfer Learning, Generative AI, Prompt Engineering*

Machine Learning Tools: *Hugging Face, Haystack, Ultralytics, DeepLabCut*

ETL Tools: *Informatica PowerCenter, Informatica Intelligent Cloud Services*

Developer Tools: *Linux, REST API, Git, GitHub, JIRA, PyCharm, Jupyter, VS Code, Eclipse, Microsoft Office Suite, Office COM Automation, Google Workspace APIs*

Infrastructure and IoT: *Linux Server Administration, ZFS, Samba, Active Directory, Raspberry Pi, MQTT, Home Assistant, RFID, Tailscale*

EXPERIENCE

Systems Engineer

Mar 2024 – Present

Hooke Laboratories, LLC

- Lead a small engineering team as technical owner of a portfolio of 30+ internal software tools that digitize and modernize laboratory and office operations, driving each project from requirements gathering and design through development, deployment, and long-term support, while enforcing version control, documentation standards, and DEV/PROD release discipline.
- Architected and developed an in-house data loss prevention (DLP) platform for outbound email, integrating REST APIs, Google Workspace APIs, and Linux-based scheduling to safeguard sensitive information and strengthen the organization's security posture.
- Built and maintain a multi-tool document operations suite adopted across the organization, leveraging Python, Office COM automation, and PyInstaller packaging with native Windows shell integration to standardize secure document workflows for business teams.

- Developed a suite of Python desktop applications supporting histology research, applying image processing, multithreaded Tkinter interfaces, and cryptographic file integrity verification, and delivered 15 iterative releases in two months in close collaboration with research stakeholders.
- Led the development of a timesheet management system built on Raspberry Pi, RFID hardware, Flask, and MariaDB, replacing manual attendance tracking and streamlining payroll processing across multiple sites.
- Designed and deployed IoT monitoring solutions for critical laboratory equipment using Raspberry Pi, MQTT, MySQL, and Home Assistant, delivering real-time alerting that protects research materials around the clock.
- Developed computer vision pipelines using DeepLabCut, deep learning pose estimation, and multi-camera 3D reconstruction to automate behavioral analysis, improving the precision, consistency, and scalability of preclinical research.
- Designed and implemented machine learning models for tissue analysis, bringing greater precision, automation, and scalability to histopathology research workflows.
- Developed and maintained large language model (LLM) pipelines that give operations teams fast, accurate access to organizational data, enabling automated responses to complex queries.
- Created software solutions to manage and control analytical instruments, optimizing workflows and improving operational efficiency for research teams.
- Engineered a broad portfolio of Python and Google Apps Script automations for research, project management, and business operations teams, integrating Google Sheets, Drive, and Gmail APIs to deliver automated alerting, reporting, scheduling, and billing workflows that significantly reduced manual effort.
- Built data quality and integrity tooling for business-critical spreadsheets and study records, combining automated validation, change detection, and anomaly flagging to improve data reliability across departments.
- Developed document processing and enterprise search tools using Apache Tika, text extraction, and pattern matching, making organizational records faster to locate and analyze.
- Built and administered Linux servers and containerized services using Docker, ZFS, and Samba with Active Directory integration, and partnered with the IT department on security protocols, data backup strategy, and day-to-day IT operations to keep the company's technology infrastructure stable and reliable.
- Serve as the primary internal consultant on artificial intelligence and machine learning, evaluating tools, providing strategic guidance, and promoting best practices to drive company-wide adoption.

Facilities Information Systems Assistant – Python Developer

Jan 2023 – Dec 2023

University of Massachusetts Lowell

Cost Data Integration and Reporting

- Developed a Python codebase using the e-builder REST API to extract cost data in CSV and XML formats, validating purchase orders and invoices and automating report generation for multiple teams.
- Optimized data import processes with iterative retrieval logic, reducing data retrieval time by 70%.
- Integrated data from multiple systems into Microsoft SQL Server, working around API record limits to enable regular retrieval for cost reconciliation, reporting, and table updates.
- Built HTML front-end interfaces and Tkinter dashboards for the work order management system, simplifying navigation and day-to-day management tasks.
- Performed unit and user acceptance testing and debugged existing code, resolving issues to improve system stability and user experience.

Pedestrian Detection System

- Led the development of a machine learning powered pedestrian detection system for the UMass Lowell Campus Planning department.
- Applied YOLO object detection with real-time tracking to detect, follow, and count pedestrians across defined regions of campus pathways.
- Built an interface for uploading videos and defining regions of interest for analysis.

- Enabled the planning team to make data-driven decisions on campus modifications, reducing manual planning time by 90%.

Programmer Analyst Trainee – Data Integration (ETL) Developer

Dec 2020 – Nov 2021

Cognizant Technology Solutions India Pvt Ltd

- Collaborated on a cross-functional team effort to enhance and optimize a Data Warehousing System, resulting in improved query performance and reduced data processing time for a Customer Relationship Management System in the Automotive Industry with a customer base exceeding 5 million.
- Designed and implemented advanced mappings and workflows using Informatica PowerCenter, streamlining ETL operations and contributing to enhanced system responsiveness.
- Oversaw the administration and maintenance of the backend Database in PostgreSQL and Amazon Redshift, implementing proactive measures that reduced system downtime by 10% and ensured data integrity.
- Delivered comprehensive system documentation, facilitating seamless knowledge transfer and equipping the team with essential resources for efficient troubleshooting and system enhancements.

PROJECTS

Evaluating Cross-domain Adaptability of Text Summarizer: News Article Summarization

Oct 2023 – Dec 2023

- Engineered advanced text summarizers, seamlessly integrating Extractive (TextRank) and Abstractive (BART) techniques.
- Enhanced BART Model performance significantly, achieving a 20% improvement over previous implementations.
- Streamlined a thorough evaluation of cross-domain adaptability, consistently outperforming benchmark ROUGE scores in contrast to the model's original implementation and a fine-tuned BBC News model.
- Analyzed the abstractive summarizer's adaptability to different domains, affirming its versatility and effectiveness.

Chest X-Ray Classification to Detect COVID-19 Using Deep Neural Networks

Feb 2023 – May 2023

- Crafted a deep-learning model for rapid and precise diagnosis of respiratory illnesses, focusing on COVID-19 detection.
- Remodeled three widely recognized deep learning architectures (ResNet, VGG, and LeNet) to train models capable of classifying chest X-ray images into four categories on 42,000 lung X-ray images.
- Achieved high accuracy in the diagnosis of respiratory illnesses, especially COVID-19 with an accuracy of 92%.
- Implemented an advanced visualization technique Grad-CAM to identify and highlight areas affected by the virus within the lungs, aiding medical professionals in targeted treatment approaches.

Climate Change Sentiment Analysis

Jan 2022 – Apr 2022

- Conducted sentiment analysis and exploratory data analysis on a dataset of over 43,000 tweets to understand public opinion on climate change using natural language processing techniques.
- Employed a Deep Learning Model with a Recurrent Neural Network approach using features including text in tweets and the frequency of specific keywords achieving an accuracy rate of 96%.

Citizens Income Prediction – Comparison of Machine Learning Models

Feb 2022 – Apr 2022

- Trained predictive models on the Adult Income Dataset, encompassing 48,000 data samples, to predict citizen income and classify individuals into two categories.
- Employed diverse machine learning algorithms, achieving an 85% overall accuracy rate and showcasing a 15% decrease in misclassifications compared to baseline models.
- Performed a thorough analysis using evaluation metrics such as Confusion Matrix, F1 score, Recall, Precision, and Accuracy, revealing the Random Forest algorithm as the most suitable for the current scenario with a 92% accuracy rate.

Face Mask Detection

Oct 2020 – Nov 2020

- Developed a sophisticated Face Mask Detection system with 90% accuracy, adapting a Convolutional Neural Network and Computer Vision techniques.

- Formulated testing on both video and photo inputs, integrating a webcam using OpenCV for real-time analysis, providing instant results above each face, and triggering a beep sound for non-compliance.

Recolored Images Detection Using Deep Discriminative Model

Jan 2020 – Apr 2020

- Coordinated the implementation of the IEEE paper which involved curating and pre-processing large datasets, fine-tuning model hyperparameters, and optimizing the network architecture for detecting fraudulent images.
- Evaluated the proposed algorithm on forged images generated by various color transfer methods and internet-collected images, achieving a 10% improvement in accuracy and demonstrating its effectiveness in real-world scenarios.

Aadhaar-Based Online Voting System

Jun 2019 – Nov 2019

- Implemented and deployed a secure and reliable alternative voting channel using Java Server Pages and SQL, contributing to a 20% increase in voter participation and 15% reduction in costs during the Student Body election.
- Ensured utmost confidentiality and accuracy, with over 2000 students casting votes through the implemented system.

CERTIFICATIONS

- Generative AI for Software Developers Specialization, *IBM*
- Generative AI for Data Scientists Specialization, *IBM*
- Introduction to C#, *Sololearn*
- Informatica PowerCenter – Beginner to Expert Level, *Udemy*
- SQL for Beginners: Learn SQL using MySQL and Database Design, *Udemy*
- Learn Python: The Complete Python Programming Course, *Udemy*
- HTML, CSS, & JavaScript – Certification Course for Beginners, *Udemy*
- Core And Advanced Java, *Sahasra SoftTech*